

Jonah Blackmon

817-724-6500 | JonahKBlackmon@gmail.com

Austin, TX | jonahblackmon.com | linkedin.com/in/jonahkblackmon | github.com/jonahblackmon

EDUCATION

The University of Texas at Austin, Austin, TX

May 2028

Bachelor of Science, Computer Science, GPA: 3.92/4.0

Relevant Coursework: Multicore Operating Systems, Operating Systems, Comp. Architecture, Algorithms, Data Structures

EXPERIENCE

Undergraduate Course Assistant

August 2025 - Present

The University of Texas at Austin | Austin, TX

- Supported instruction for 400+ students through office hours and independently led classes of 30+ students
- Assisted course instruction for Discrete Mathematics and Elements of Software Design, covering proofs and OOP

Undergraduate Student Researcher

January 2025 - Dec 2025

Living with Robots Lab, Texas Robotics | Austin, TX

- Research in autonomous intelligent robotics with a focus on real-time human-robot interaction
- Built a real-time gesture-aware multimodal LLM using discrete motion tokenization for conversational interaction

Software Engineer

September 2025 - Present

Longhorn Developers | Austin, TX

- Developer fellow for the Longhorn Developers UT Mobilize team at the University of Texas at Austin
- Developing a mobile app to assist UT students with disabilities in navigating campus using accessible routing

Information Technology Intern

May 2025 - August 2025

IT Integrations | Fort Worth, TX

- Executed network overhauls at healthcare facilities, including system reconfiguration and device migration
- Troubleshoot end-to-end network failures on site during live network transitions, working independently

PROJECTS

Multicore Distributed Operating System | C++, Assembly, Linux, Rust, QEMU, Networking

- Developed kernel functionality for a multicore operating system supporting user-space program execution
- Implemented POSIX-style signal delivery/handling, a preemptive multilevel feedback queue (MLFQ) scheduler, and coroutine infrastructure in a 30+ person codebase with structured design reviews and feedback integration

CampusNav - Full-Stack Navigation App | Swift, Git, Firebase, MapBox, MapKit, SwiftUI

- Published a navigation app specifically for college campuses to improve the student experience of getting to class
- Implemented a custom A* routing engine over a complete graph of entrances/pathways, leveraging existing databases, optimizing routes through buildings and pedestrian shortcuts, reducing average walking time by 30%

Gesture-Enhanced Semantic Tokenization Architecture (GESTA) | Python, ROS2, PyTorch, Git

- Developed an architecture that tokenizes continuous hand motion using an unsupervised VQ-VAE
- Enables dynamic gesture learning, treating motion primitives as language-level tokens for multimodal reasoning

CarGaze - HackTX Best Pitch Winner, Fall 2025 | Swift, Supabase, Google Gemini, SwiftUI

- Led full-stack development of a mobile app designed for the Toyota Financial Services track, helping users discover vehicles aligned with their budget and financing options
- Integrated a Gemini-based recommendation engine for real-time affordability insights, earning *Best Pitch* among 1,000+ participants for innovation and financial accessibility

LEADERSHIP & COMMUNITY INVOLVEMENT

ECLAIR Robotics, Austin, TX

September 2024 - Present

Member, C.R.A.C.K.L.E. (Contextual Robotics Arm for Knowledge Learning and Exploration)

- Worked on gesture detection using Leap Motion and integrated ML algorithms for contextual robotic response

Levinson Foundation, Santa Fe, NM

June 2023 - Present

Next-Gen Board Member

- Appointed to youth board researching and funding activism initiatives; responsible for the disbursement of \$100,000 of funding to organizations, as well as attending national conferences and climate advocacy events

SKILLS

Languages: Java, Python, Swift, C, C++, JavaScript, HTML, CSS, Assembly

Tools & Frameworks: React, SwiftUI, ROS2, PyTorch, Azure, Firebase, Git, Linux, OpenCV, AVFoundation, MapBox

Related Skills: CAD, Blender, Adobe Photoshop, Adobe Illustrator, Adobe InDesign, Adobe After Effects